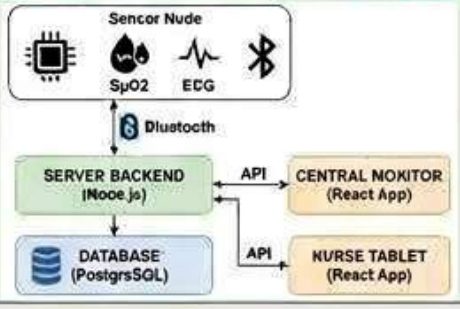
Project based learning - FSWD

HEALTHPULSE: REAL-TIME PATIENT TELEMETRY AND ICU MONITORING PLATFORM

ABSTRACT

HealthPulse is a full-stack platform designed for real-time patient data telemetry from sensors (like Heart Rate, SpO2, Temperature) directly to an ICU central monitor and mobile app. Built with PostgreSQL, Node.js/Express, and React, it enables instantaneous, non-invasive vital signs tracking and neutral visualization for entire wards. Goal is faster clinical response and better patient outcomes.

SYSTEM ARCHITECTURE



CO-1

React - API Service Layer



Fetching Patient Data

Fetching Patient Data

CO-2

Node.js - Express Server Setup

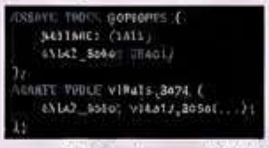


API and Database Connection

API and Database Connection

CO-3

PostgreSQL - Database Schema



Patients and Vitals Tables

Patients and Vitals Tables

CO-4

React - Vitals Component

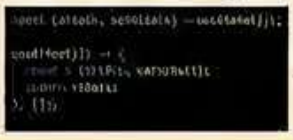


Vitals Display Logic

Vitals Display Logic

CO-5

React - State Management (Hooks)



Managing Vitals Data

CO-6

React Entry Point / Vite Build



React App Entry Point

Vite Bulid Configuration

CONCLUSION

The HealthPulse Telemetry project successfully demonstrates the development of a real-time ICU monitoring platform using full-stack engineering with PostgreSQL, Node.js, and React. It achieves reliable data acquisition, processing, and visualization. Through the practical implementation of Course Outcomes (CO1–CO6), the project results in a scalable, performant, and user-friendly system, enabling enhanced patient care.

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